Introduction to Computer Science (CSL1010)

Prelim – L1

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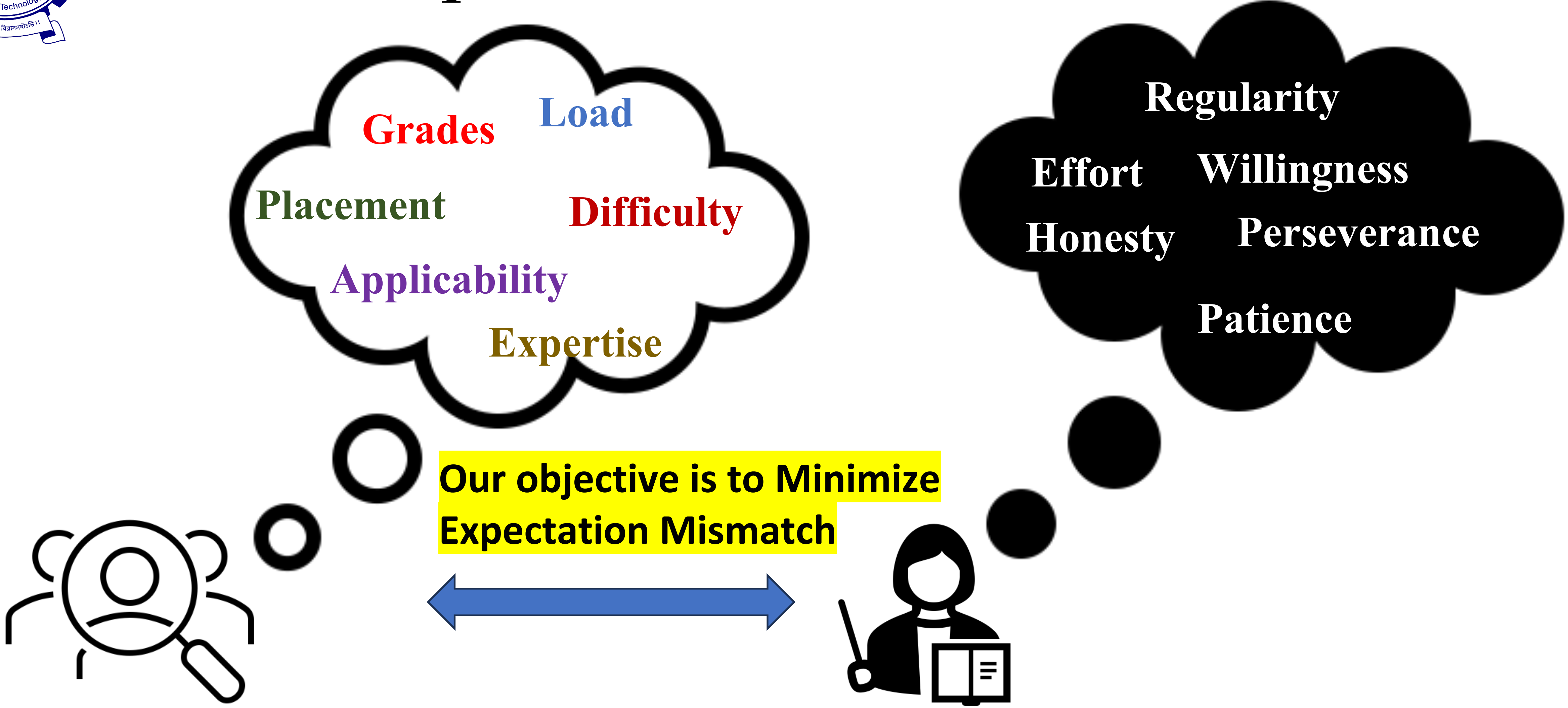
Why study ICS?

- Fulfill BTech Requirement…
- Job Prospect…NOT just IT btw!
- Want to learn Programming…
- Curious about Computer…

Course Expectations



Our objective is to Minimize Expectation Mismatch

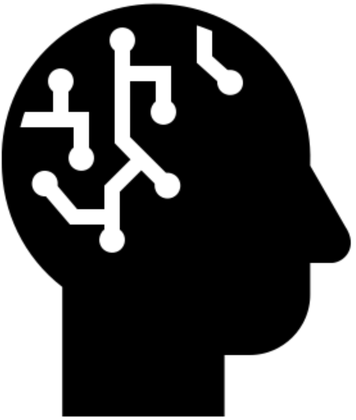
How many of you have NOT used ChatGPT so far?

https://www.outrightcrm.com/blog/will-chatgpt-replace-coders/

Will ChatGPT Replace Coders?

- Generates programs in multiple languages
- Debugging becomes easy
- Faster response time
- Automate repetitive tasks
- break down complex logic

- Contextual Reasoning
- Critical and strategic thinking
- Adapt decisions as per constrained dynamics
- Collaboration
- Fixing errors based on perception and expertise
- Develop Extendable, Sustainable and Scalable system



Real Coders in Making!!

Experience from Failure matters

What we want **Human Experts** *with* **Computers**

So, What are we going to learn?

The Fundamentals:

- Organization
- Operating Systems
- Computational Thinking
  - Problem solving
  - Systems design
  - Algorithms

Programming:

- Basics of data representation
- Programming constructs
- Object oriented programming
- Programming paradigms

"Computer science is the study of computers and computational systems, focusing on the **theoretical and practical** foundations of how information is processed, stored, and communicated".

Assessment

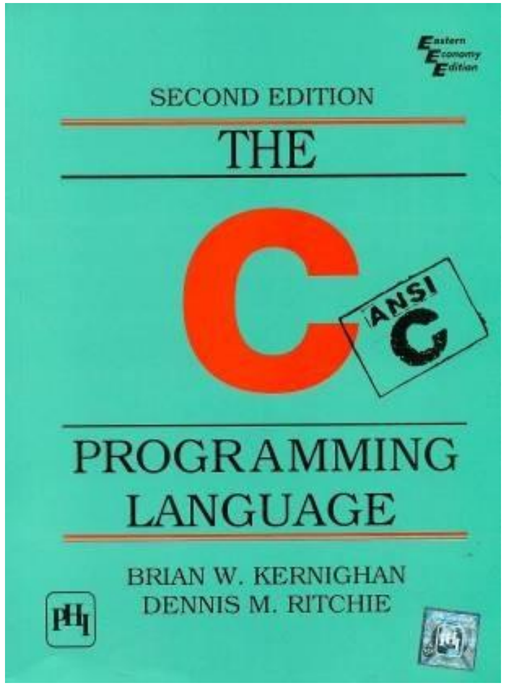
Continuous Evaluations: 40%

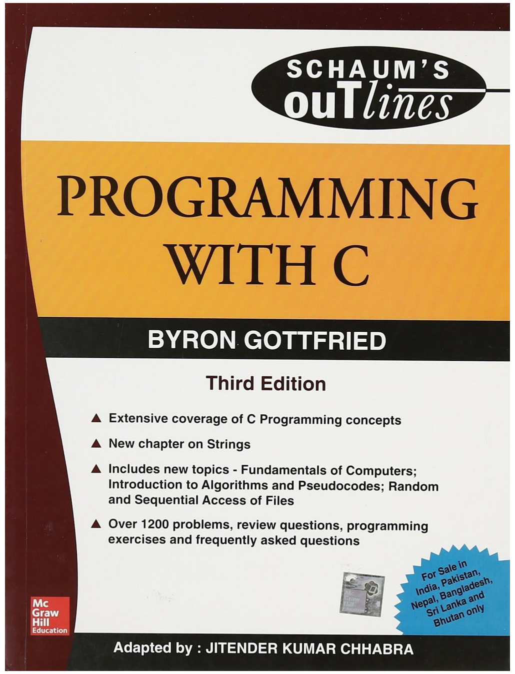
- Assignments, Lab, Attendance and CP, Quizzes
- Project & viva

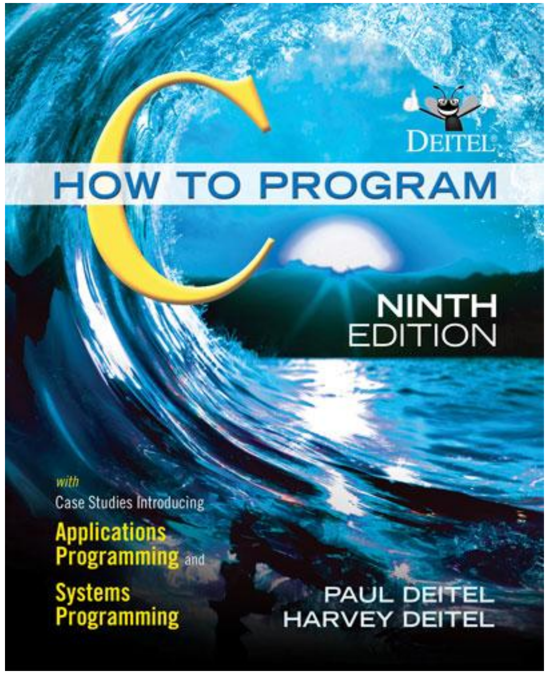
Primary Evaluations: 60%

- Minor: 20%
- Major : 40%

Books


Eastern Economy Edition
SECOND EDITION
THE
C
ANSI C
PROGRAMMING
LANGUAGE
BRIAN W. KERNIGHAN
DENNIS M. RITCHIE
PHI


SCHAUM'S ouTlines
PROGRAMMING
WITH C
BYRON GOTTFRIED
Third Edition
Extensive coverage of C Programming concepts
New chapter on Strings
Includes new topics - Fundamentals of Computers; Introduction to Algorithms and Pseudocodes; Random and Sequential Access of Files
Over 1200 problems, review questions, programming exercises and frequently asked questions
For Sale in India, Pakistan, Nepal, Bangladesh, Sri Lanka and Bhutan only
Mc Graw Hill Education
Adapted by : JITENDER KUMAR CHHABRA


C
DEITEL
HOW TO PROGRAM
NINTH
EDITION
with
Case Studies Introducing
Applications
Programming and
Systems
Programming
PAUL DEITEL
HARVEY DEITEL

Course Logistics

English language offering:

- Slot-G (Mon: 5-6 PM; Wed, Fri: 1-2 PM)

Non-English language offering:

- Slot-K (Mon, Wed, Fri: 8-9 AM)

- Google Classroom Code: **erdrzsoc**

- Lab Slot: Mon [11-1], Tues [11-1], Wed [9-11]

- Tutors: Prof. Richa Singh, Prof. Mayank Vatsa, Prof. Bikash Santra, Prof. Manish Agarwal

- TAs: Details will be intimated over GC

- Office Hours: Mon-Fri [11:00-13:00]

Range of Computers

As of late 2025, the U.S. Department of Energy's **El Capitan** is the world's fastest supercomputer, achieving a performance of **1.809 exaFLOPS** (1,809 petaFLOPS)

India's current fastest supercomputer, **AIRAWAT-PSAI** (installed at C-DAC Pune), delivers a peak performance of **13,170 teraflops (13.17 Petaflops)**

Mainframe and supercomputers

General purpose computers – PC, Laptops

Range of embedded computers

Size **Purpose** **Capacity** **Cost** **Technology**

Evolution of Computer

Lets take the lid off a desktop computer

PROCESSOR

- (Under the heatsink)

MOTHERBOARD

- With ports

GRAPHICS CARD

POWER SUPPLY

- Converts electricity so it can be used by the components

MEMORY

- RAM

STORAGE

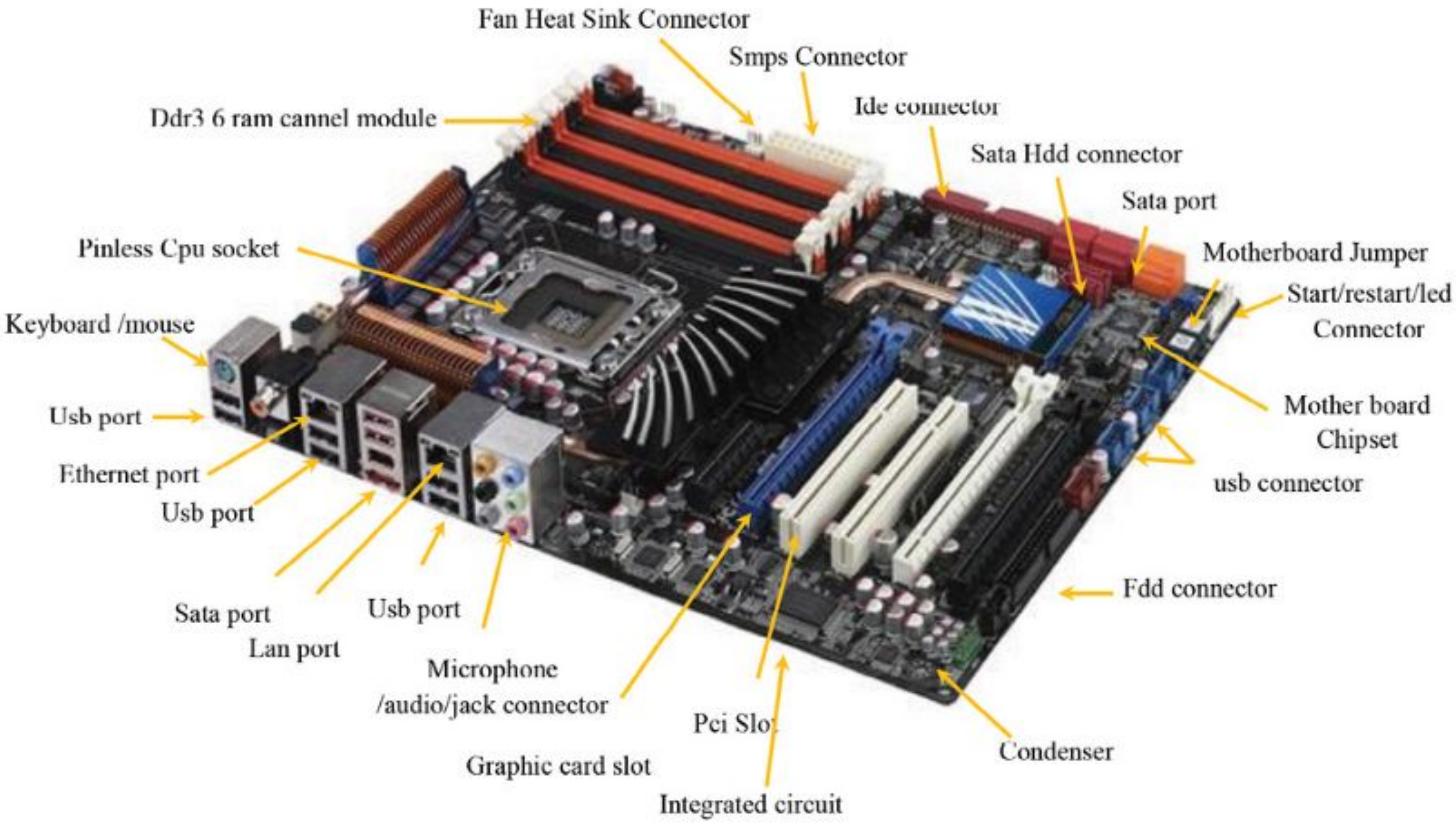
- (Optical drive)

STORAGE

- Hard drive

An inside peek at a modern motherboard

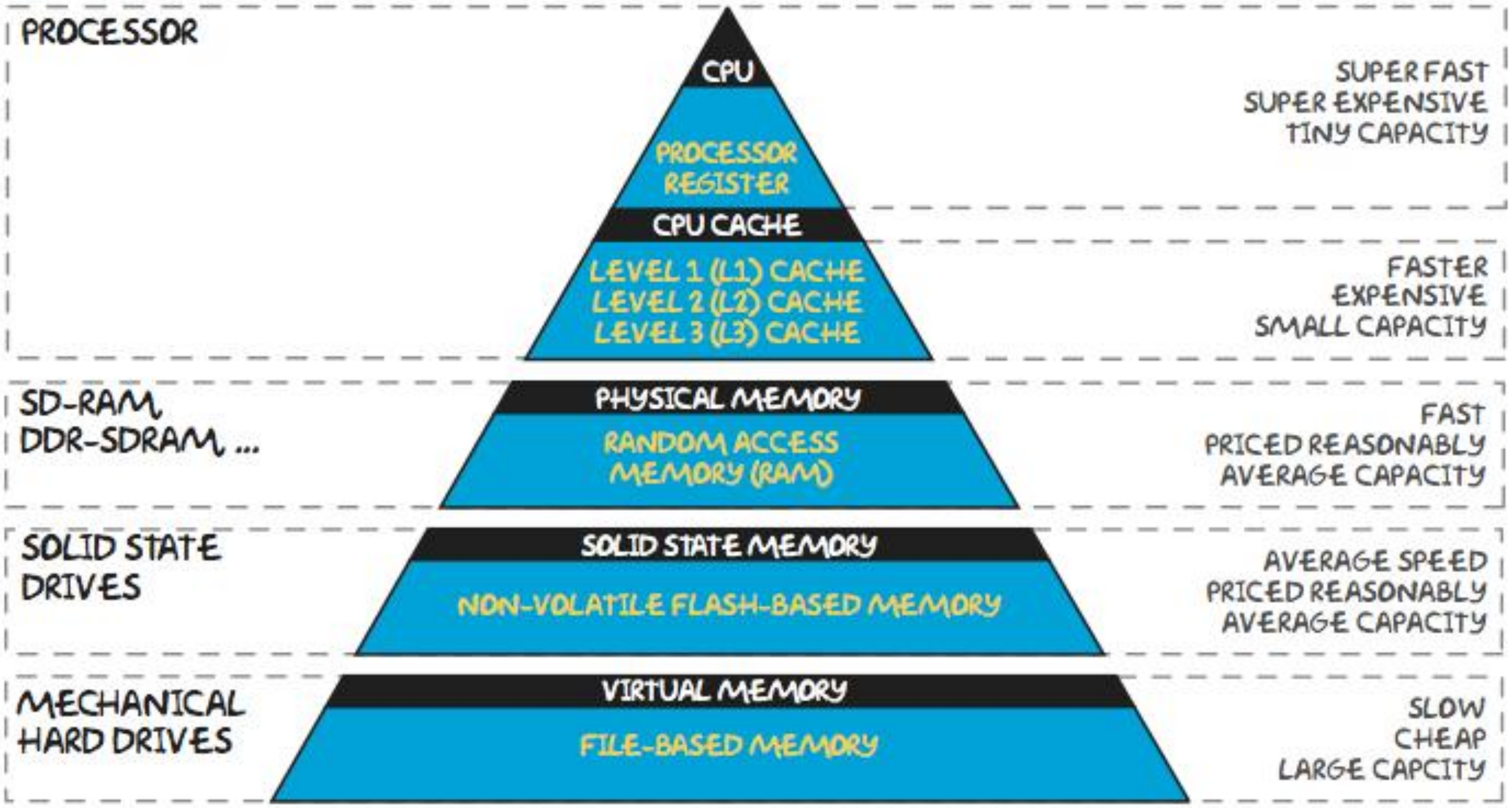
Watch: https://youtu.be/bR-DOeAm-PQ



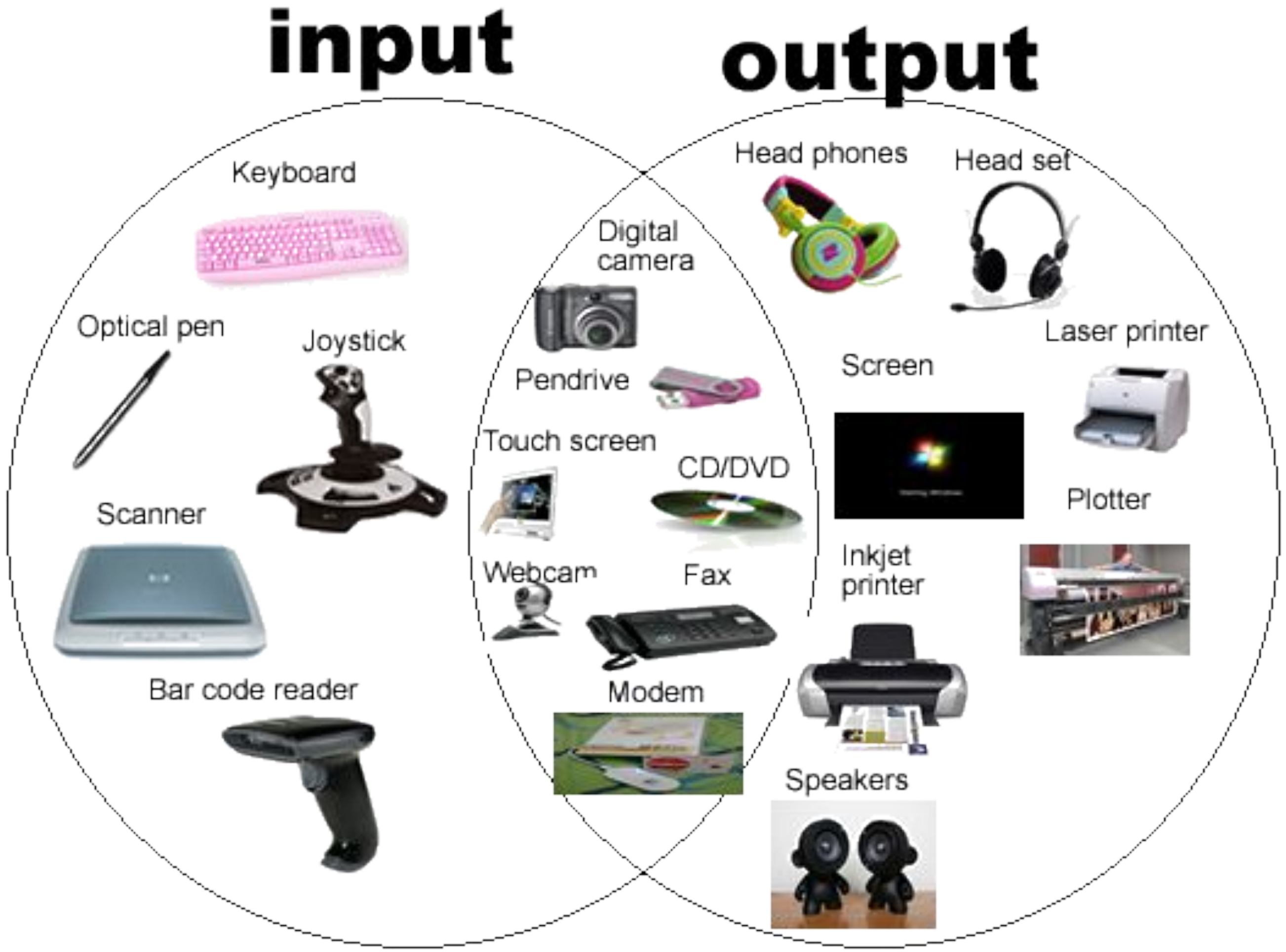
source: https://computerscience.chemeketa.edu/cs160Reader/ComputerArchitecture/MemoryHeirarchy.html

Memory: Storage Hierarchy

Watch: https://youtu.be/r3Jy5dHOj3g

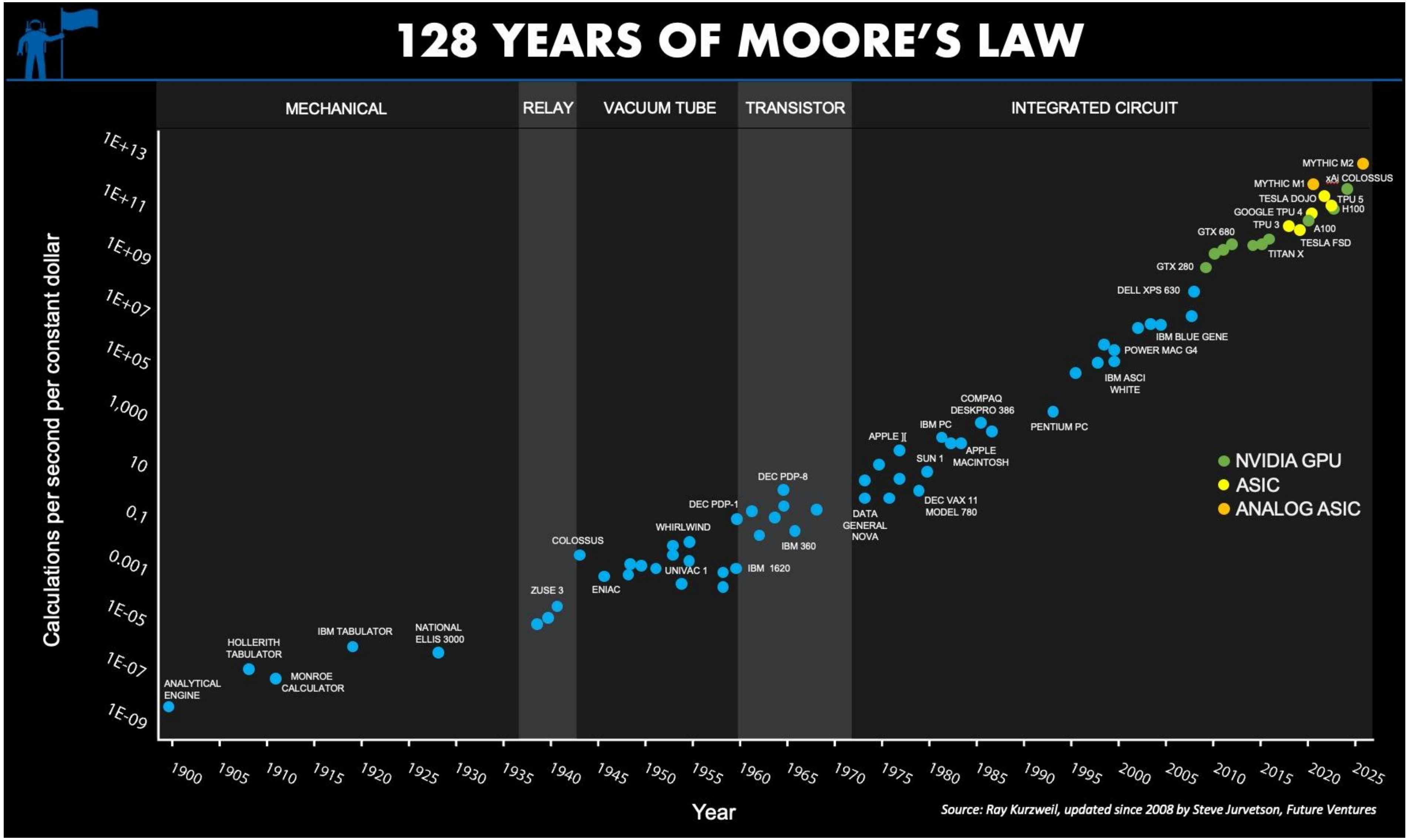


Range of Inputs and Outputs



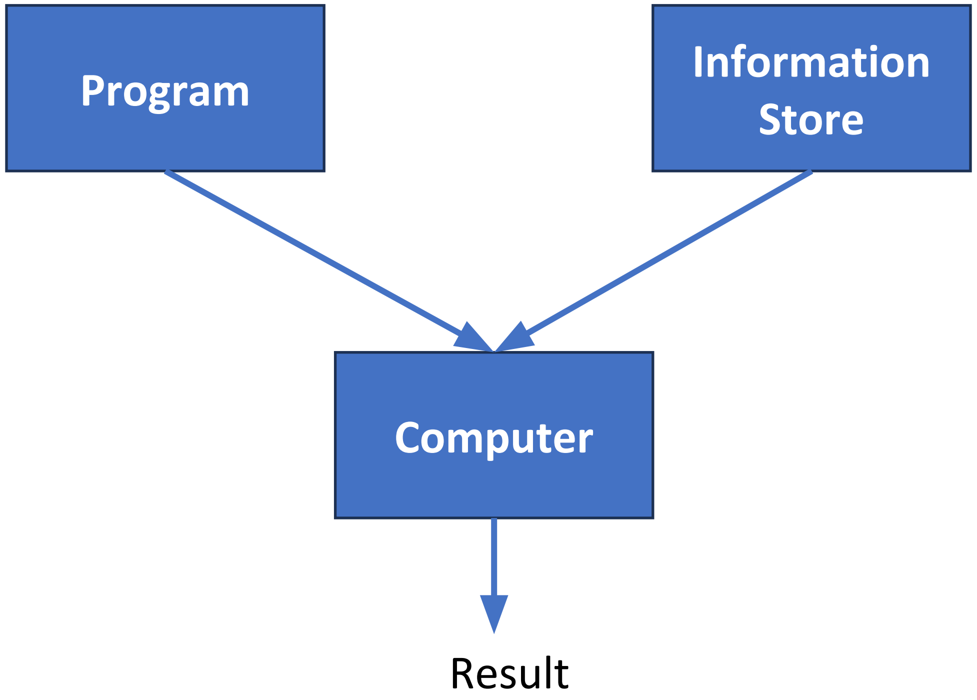
https://en.wikipedia.org/wiki/Moore%27s_law

Moore's Law is the observation that the number of transistors on a microchip doubles approximately every two years, leading to increased computing power, smaller devices, and lower costs. Originally a prediction made by [Gordon Moore, co-founder of Intel, in 1965]

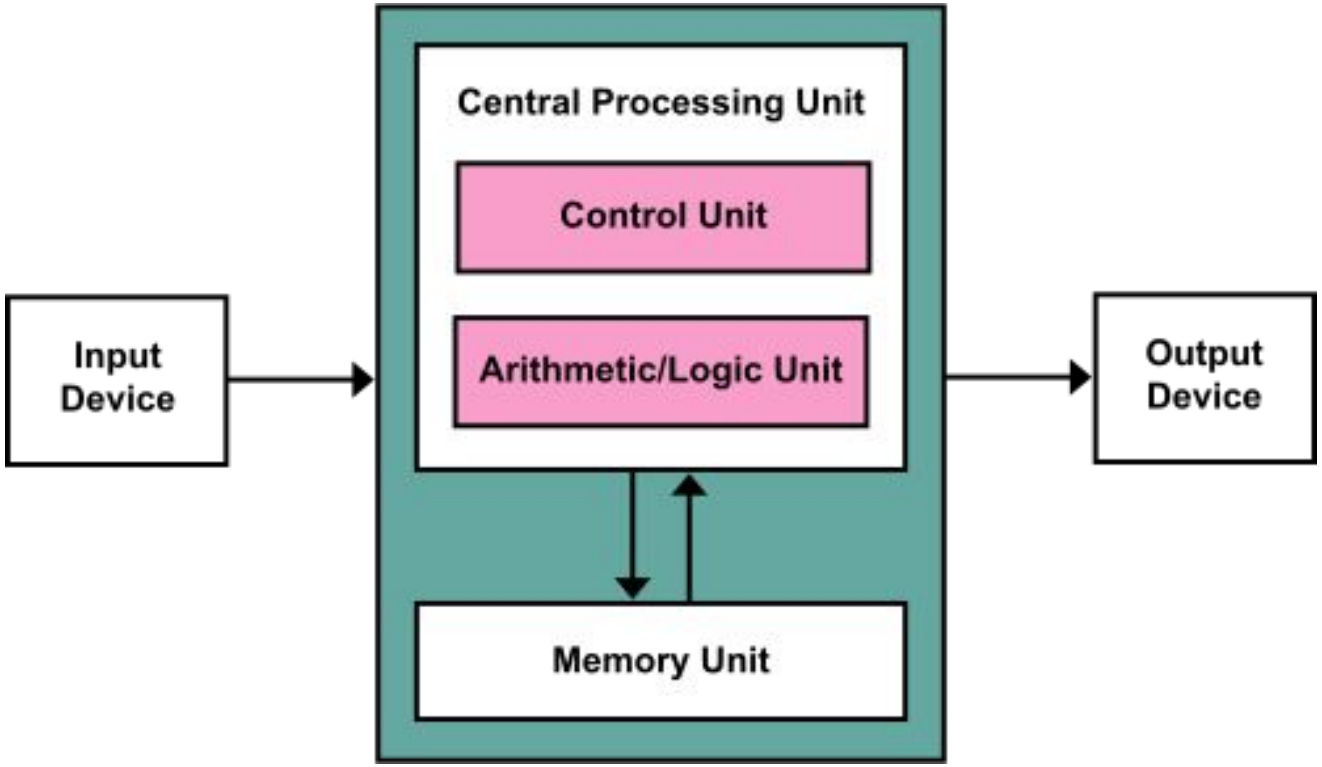


How does it work?

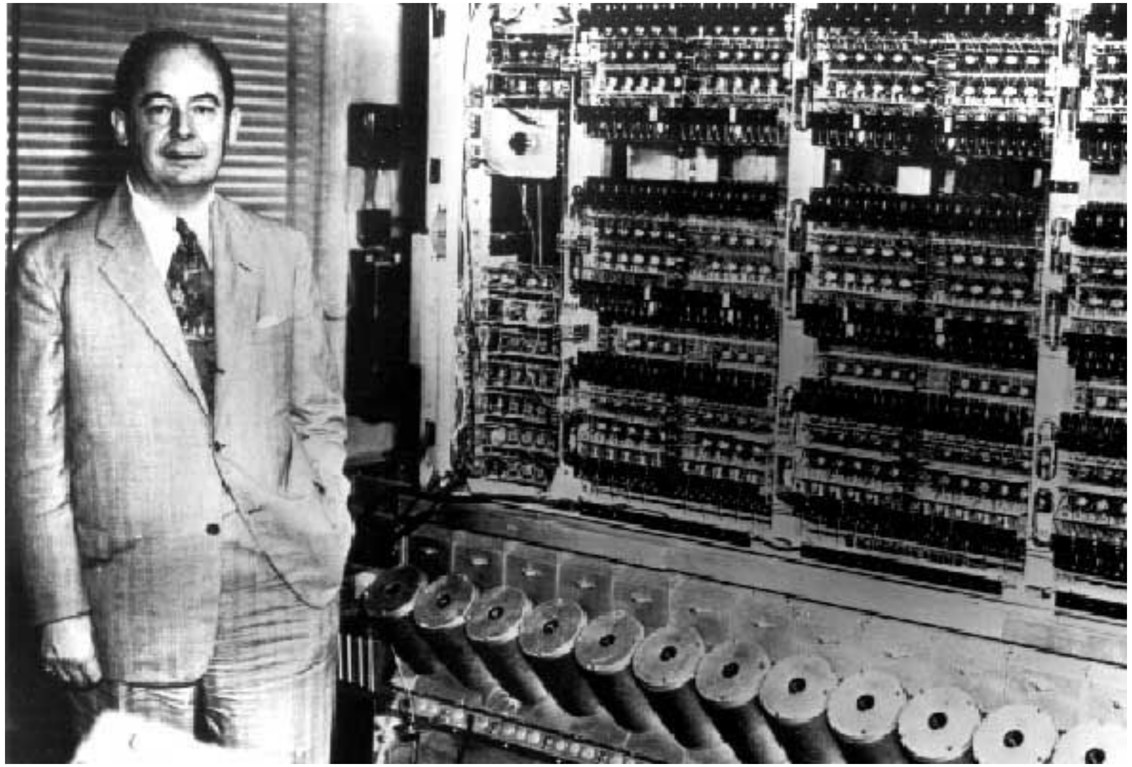
- Program – list of instructions given to the computer
- Information store – data in the form of files
- Computer – process the information store according to the instructions in the program



Von Neumann Architecture

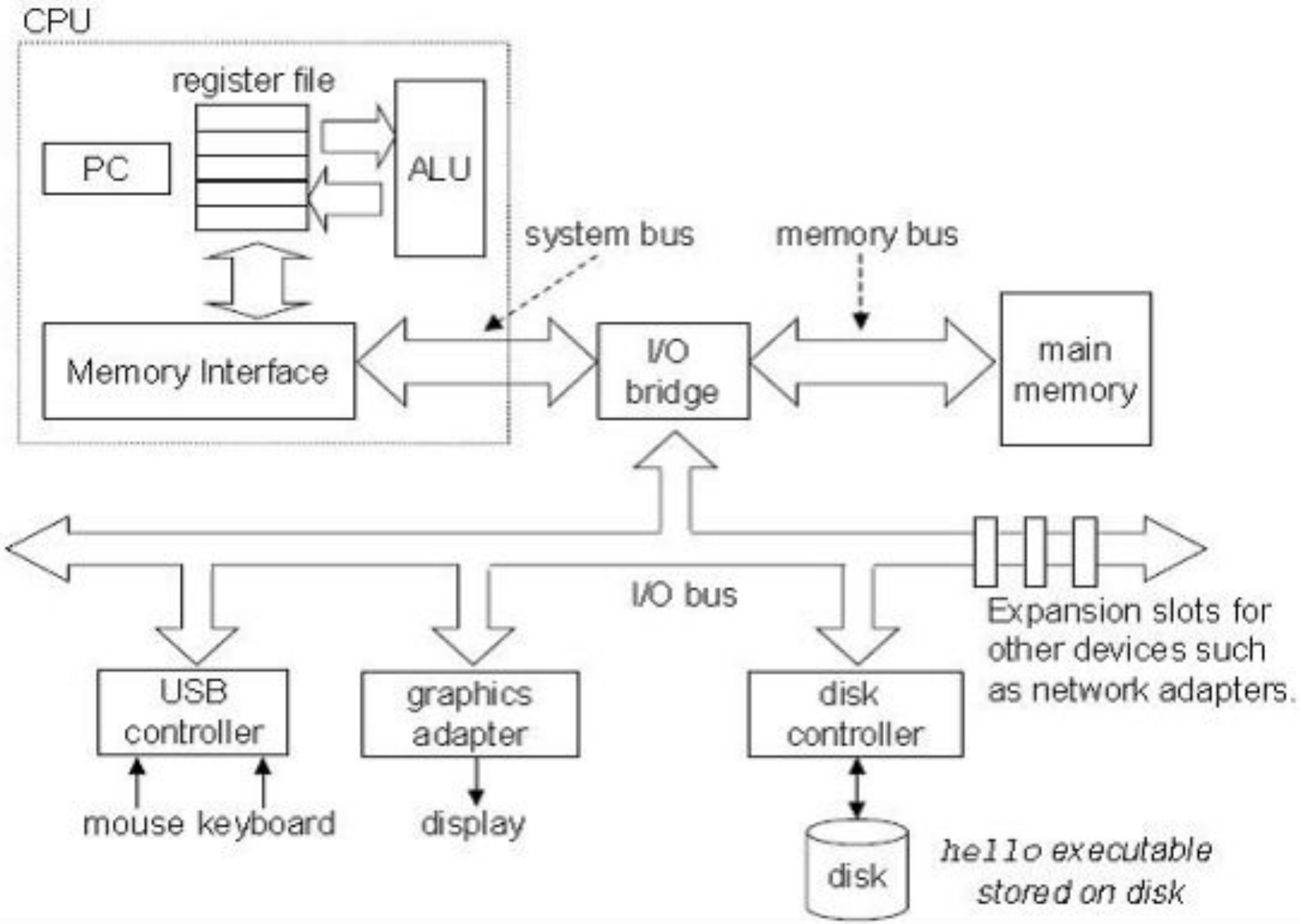


source: wikipedia



Source: http://www.gap-system.org/~history/PictDisplay/Von_Neumann.html source: http://www.iluvex.com/reading/2015/08/01/Reread-Computer-Systems_1

Computer Hardware Organization



When Human Users interact with Machines

Interaction

Users

Computer

01001010101
10110010101
10001110100
11001010101
11010101010

The language that human users can interpret

The language that computers can interpret

https://www.learncomputerscienceonline.com/instruction-set-architecture/

Instruction Set Architecture (ISA)

- A well-defined hardware software interface
- Functional definitions of storage locations and operations
- Precise description of how to invoke and access them
- Advantage: *different implementations (cost, performance, power) of the same architecture*
- Disadvantage: *sometimes prevents using new innovations*
- Common instruction set architectures:
  - IA-32, PowerPC, MIPS, SPARC, ARM, and others

ISA Includes

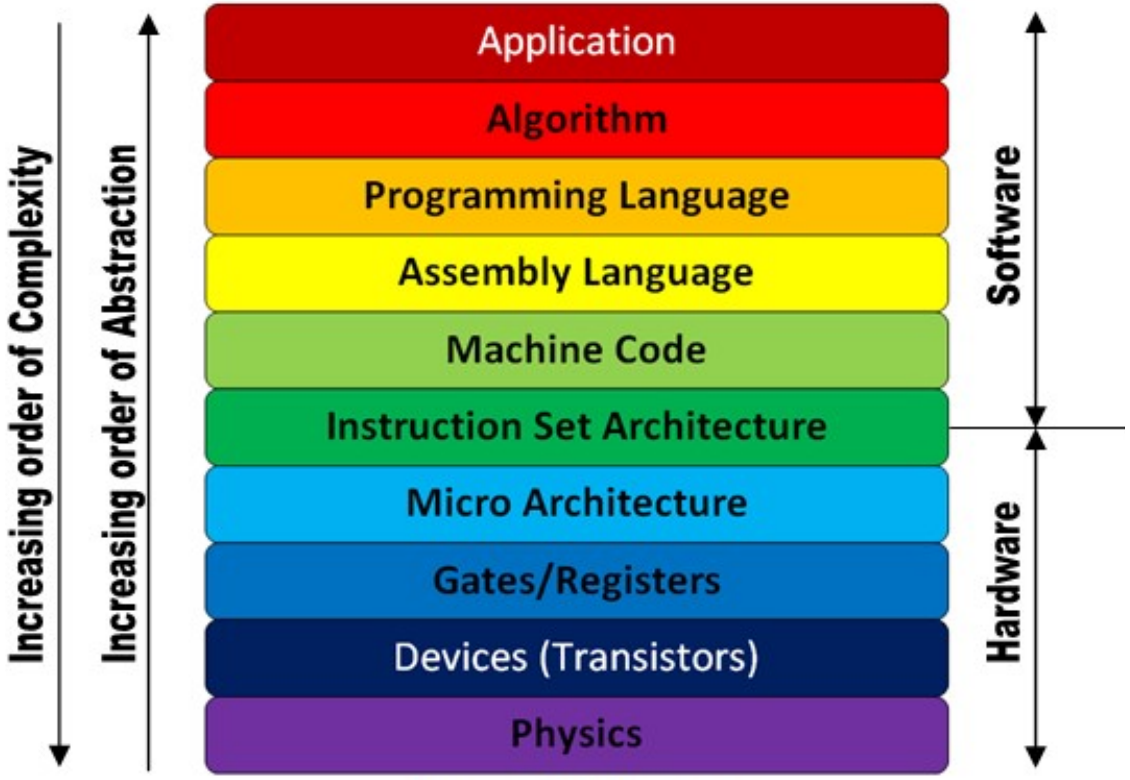
- Organization of storage
- Data types
- Encoding and representing instructions
- Instruction Set (or opcodes)
- Modes of addressing data items/instructions
  - Arithmetic instructions : add, sub, mul, div
  - Logical instructions : and, or, not
  - Data transfer/movement instructions
- Program visible exception handling

Problems with writing programs using ISA

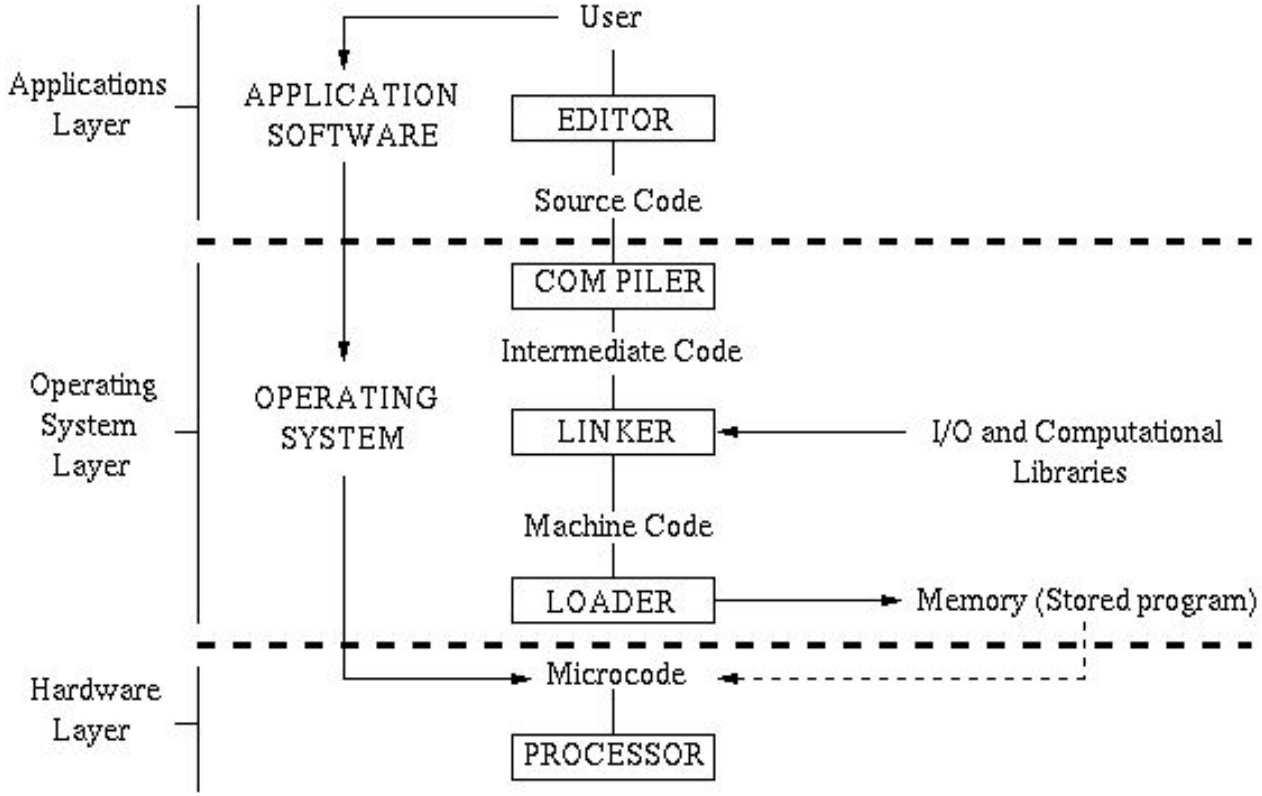
- Instruction sets of different types of CPUs are different
  - Need to write different programs for computers with different types of CPUs even to do the same thing
- **Solution**: High level languages (C, C++, Java,…)
  - CPU neutral, one program for many
  - Compiler to convert from high-level program to low level program that CPU understands
  - Easy to write programs

source: https://www.cise.ufl.edu/~mssz/CompOrg/AppxB-CompOrg.html

Computer System: A Layered Abstraction

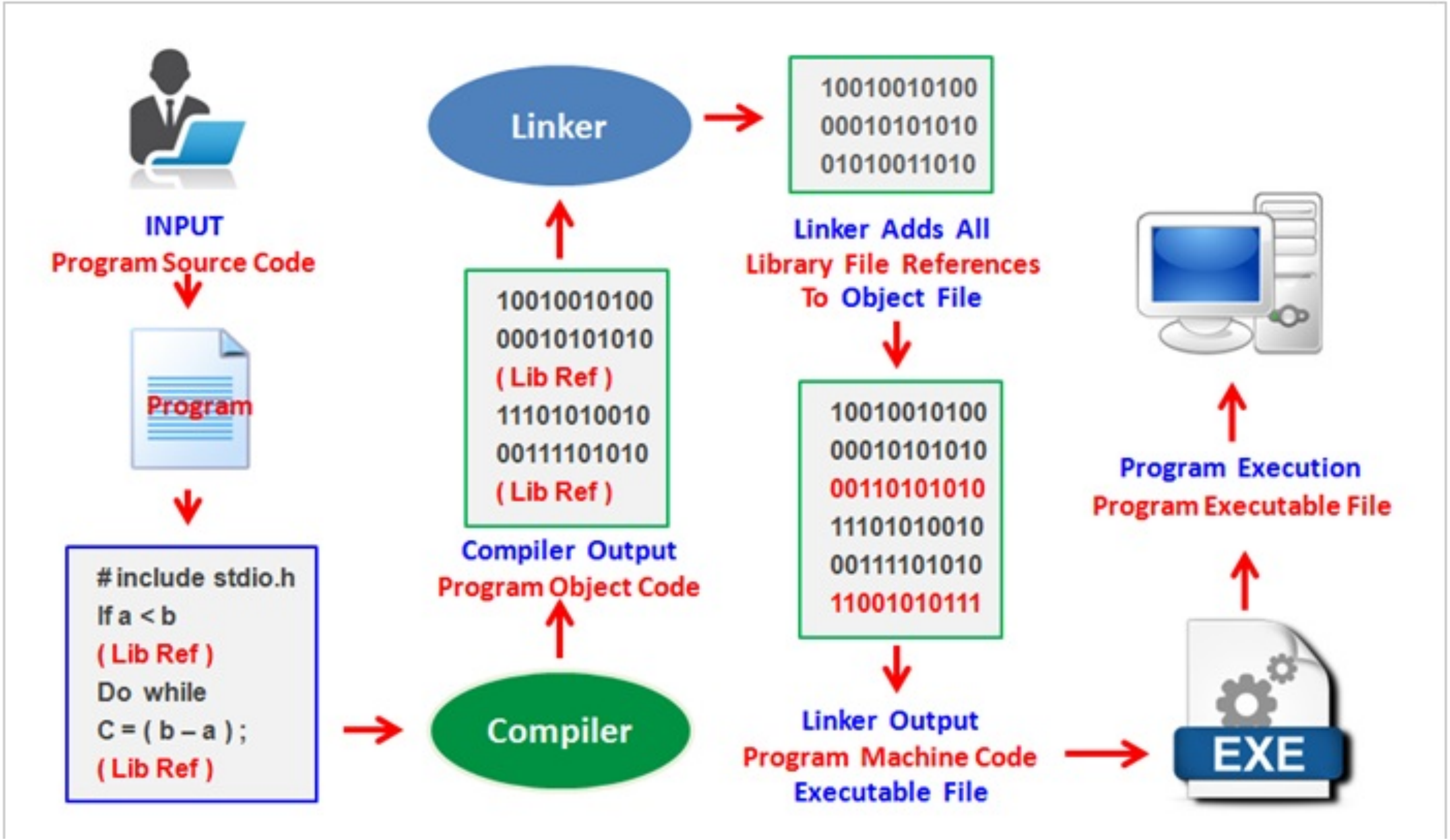


Layers of Abstraction

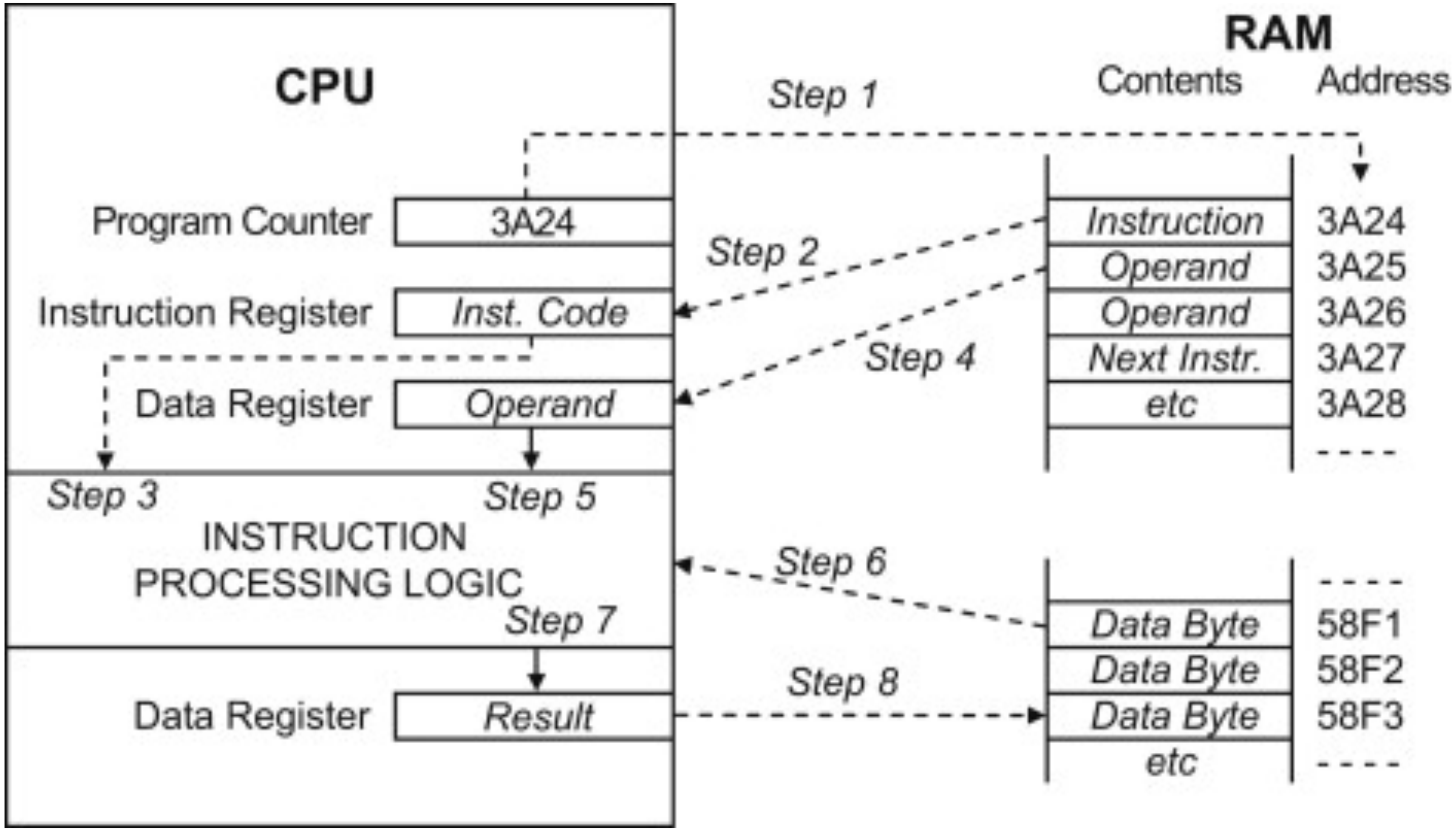


source: https://www.learncomputerscienceonline.com/machine-instruction/

Step by Step Program Execution

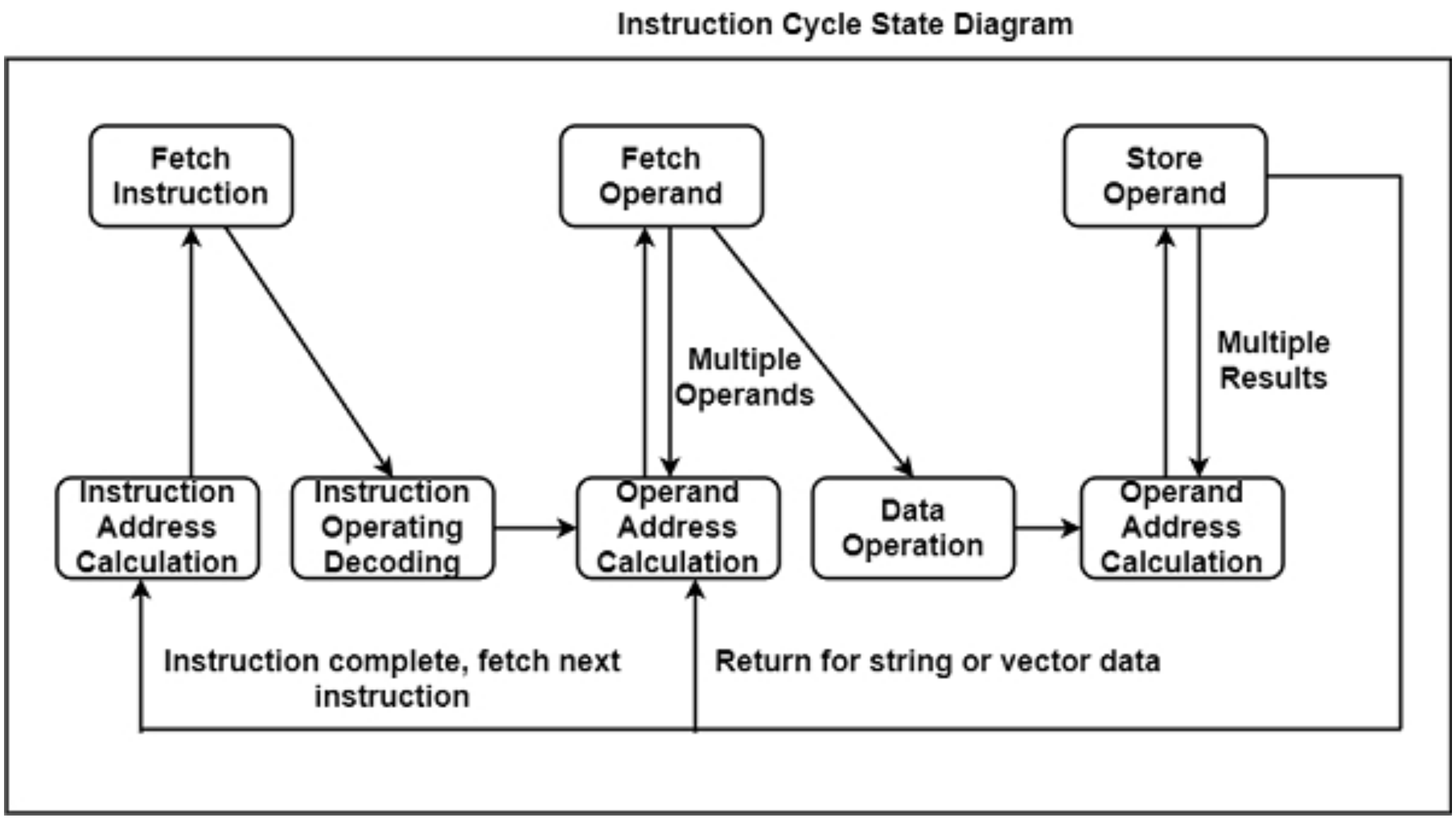


Instruction Execution Flow



https://www.tutorialspoint.com/what-is-instruction-cycle-in-computer-architecture

Instruction Cycle in Nutshell



Introduction to Computing

- You don’t need a computer for Computing
- Model of computation: To choose appropriate primitives to describe an algorithm (e.g. abacus, scale, compass, calculator etc.)
- With each of these models of computing, the rules for specifying a solution (algorithms) are different, and the precision of the solution also differs.
- **Algorithm:** a finite specification of the solution to a given problem (definite input and output). It is unambiguous and specifies a solution in terms of a finite process (finite number of steps in execution).
- **Effective Algorithm:** When the solution is specified according to a model of computation (particular primitives).
- **Program:** Encoding of an effective algorithm in the syntax of a programming language. This is necessary for machine execution of an algorithm.